

BYEONGHWA JEONG

Ph.D. AT GEOGRAPHY (GEOSPATIAL DATA SCIENCES)

R&D Associate, Geospatial Science and Human Security Division, Oak Ridge National Laboratory,

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EDUCATION

University College London

London, UK

Ph.D. in Geography, Department of Geography

Sep 2019-Jun 2023

- Thesis: "An Integrated Deep Learning Model using Retail, Residential and Footfall Changes data to Predict Retail Gentrification"
- Supervisor: Prof Paul Longley and Prof James Cheshire
- Developed a bespoke AI model (Autoencoder-LSTM) to predict gentrified high streets across England by analysing longitudinal changes in residential, retail, and footfall variables.
- Estimated the long-term impact of the COVID-19 pandemic on retail gentrification using Multivariate Multiple Regression and a Likelihood Ratio Test, finding that its effects were insignificant.
- Supervisors: Professor Paul Longley and Professor James Cheshire

The University of Edinburgh

Edinburgh, UK

Master of Science in Geographical Information Sciences, School of Geosciences

Sep 2017-Aug 2018

- Dissertation: Anticipation of Commercial Gentrification through Time and Space Seoul, South Korea (Distinction).
- Predicted retail gentrification in Seoul using a Self-Organising Map (SOM) and Cellular Automata Markov model with land price, retailer, and socioeconomic data.

Kyungpook National University

Daegu, South Korea

Bachelor of Arts in Geography

Mar 2008-Aug 2015

RESEARCH AND PROFESSIONAL EXPERIENCE

Oak Ridge National Laboratory

Knoxville, USA

R&D Associate

Feb 2025-present

- Participating LandScan Global and Mosaic projects that generate 1km and 100m global gridded population data
- Integrating satellite-based damage assessments with population models to generate near real-time estimates of populations affected by disasters and conflicts
- Developing a global habitable probability mask using AI to improve the accuracy of gridded population data
- Building an AI model to predict building-level wealth indices from multi-modal data (satellite images, building features and building footprint data) to provide critical insights for urban inequality and policy
- Applying multiple instance learning for the survey-based data to produce more reliable small-area estimates of household wealth distribution.

The University of Toronto

Toronto, Canada

Post-Doctoral Research Fellow

Jul 2023-Jan 2025

- Participated in a downtown recovery project that measure the degree of recovery of downtown areas from the COVID-19 pandemic.
- Estimated the uncertainty of sample footfall data.
- Analysed the recovery rates of individual buildings located in downtown areas and investigated the relationship between building characteristics and recovery rates.
- Imputed missing footfall data using deep learning-based imputation models.
- Developed new downtown boundaries using unsupervised image segmentation models and open-source spatial big data.
- Explored the influence of socioeconomic variables on downtown recovery rates through explainable AI.

Daejeon Sejong Research Institute

Daejeon, South Korea

External Collaborative Researcher

Oct 2024-Dec 2024

- Participated in a research project "Big Data-Driven Strategies for Enhancing the 'Science City Daejeon' Brand: Focusing on the 2024 Korea Science Festival and Daejeon Science Festival".
- Crawled web contents about Korea Science Festival and Daejeon Science Festival between 2019 and 2024
- Conducted preprocessing of the collected text data and performed text mining through TF-IDF, Latent Dirichlet Allocation and network analysis.
- Explored changes in people's perceptions of science festivals.

Daejeon Sejong Research Institute

External Collaborative Researcher

Daejeon, South Korea

Aug 2023-Oct 2023

- Participated in a research project "Discover 'Regional Things' using Big Data Analysis and Civic Collaboration".
- Crawled text data from the diverse internet sites.
- Conducted preprocessing of the collected text data and performed text mining through TF-IDF, Latent Dirichlet Allocation and network analysis.

Daejeon Sejong Research Institute

External Technical Consultant

Daejeon, South Korea

Jul 2022-Sep 2022

- Participated in a research project "An Analysis of the Concept of Hip and Hot Place in Daejeon Using Social Media Data and Text Mining".
- Advised how to crawl text data on the internet using Python and various packages
- Technical consulting to conduct preprocessing for the crawled text data for the text mining
- Consulted how to perform text mining, particularly Latent Dirichlet Allocation and network analysis for text data
- Overall advice to conduct network analysis for text analyses

Korea Environment Institute

Part-time modeler

Sejong, South Korea

Aug 2021-Oct 2021

- Built a spatial interpolation model based on deep learning in order to more accurately predict unobserved areas by Orbiting Carbon Observatory-2 (OCO-2) satellite.
- The model interpolated unobserved areas by combining CO2 modeling (GEOS-Chem) and OCO-2 data.
- This model achieved superior results than traditional interpolation methods.

Korea Research Institute for Human Settlements

Assistant Research Fellow

Sejong, South Korea

Sep 2018-Sep 2019

- Participated in a study on the usage of national territory monitoring indicators for policymakers. Participated in a study on the analysis and application of the database of national territory monitoring indicators.
- Built accessibility indicators to daily infrastructures (schools, parks, etc.) based on a grid-based map covering South Korea.
- Created an old building proportion data for the grid-based map using a building dataset recording the trajectory of all buildings in South Korea

Kyungpook National University

Student Research Assistant

Daegu, South Korea

Nov 2015-Jun 2016

- Took part in "Evaluating sensor signature changing trends of GHG satellite caused by UN forestry carbon project"
- Based on the Orbiting Carbon Observatory-2 satellite data, explored the distribution of carbon in urban, in the case of Seoul.
- Learned the basic techniques of python to preprocess satellite data, also basic geostatistic theories and applications.

PUBLICATIONS**Peer-reviewed journal publications (*published*)**

- [P10] Bokyeong Lee; **Byeonghwa Jeong (corr. author)**, Development of Grid-Based Population Projection Method: A Modified Cohort Component Approach Applied to South Korea. *Population, Space and Place* 32, 1 (2025). <https://doi.org/10.1002/psp.70161>
- [P9] Xiaoxin Zhou; **Byeonghwa Jeong (corr. author)**; Karen Chapple, A Large Language Model-based Chatbot System Framework for Urban Planners, *Expert Systems With Applications* 130307 (2025). <https://doi.org/10.1016/j.eswa.2025.130307>
- [P8] **Byeonghwa Jeong**; Bokyeong Lee. Popnet: computer vision based bespoke deep learning model for forecasting gridded population. *International Journal of Geographical Information* 0, 0, 1–20 (2025). <https://doi.org/10.1080/13658816.2025.2514792>
- [P7] **Byeonghwa Jeong**; Juhwan Song; Younghoon Kim. A first open retail boundary dataset for South Korea using open data and computer vision technique. *Scientific Data* 12, 622 (2025). <https://doi.org/10.1038/s41597-025-04958-1>
- [P6] **Byeonghwa Jeong**; Karen Chapple. Does Class Matter? Understanding Differential Pandemic Recovery via a Building Typology. *Environment and Planning B: Urban Analytics and City Science* 0, 0 (2025). <https://doi.org/10.1177/23998083251326128>
- [P5] Amir Forouhar; Karen Chapple; Jeff Allen; **Byeonghwa Jeong**; Julia Greenberg. Assessing Downtown Recovery Rates and Determinants in North American Cities After the COVID-19 Pandemic. *Urban Studies* 62, 6, 1209-1231 (2024). <https://doi.org/10.1177/00420980241270987>
- [P4] **Byeonghwa Jeong**; Jeff Allen; Karen Chapple. A new commercial boundary dataset for metropolitan areas in the USA and Canada, built from open data. *Scientific Data* 11, 422 (2024). <https://doi.org/10.1038/s41597-024-03275-3>
- [P3] **Byeonghwa Jeong**; Junwoo, Kim. Topic Modeling of the Transition of the Sense of Place in Urban Regeneration Area: A

Case Study of Daegu Bangcheon Market. *Journal of Daegu Gyeongbuk Studies* 19, 1 (2020):27-44

- [P2] **Byeonghwa Jeong**; Junwoo, Kim. Application and Evaluation of Self-Organising Map-based Spatial Clustering for Regional Policy Recommendations. *Journal of the Korean Geographical Society* 54, 3 (2019):387-404.
- [P1] **Byeonghwa Jeong**; Jung-Sup Um. Comparative evaluation of South versus North Goseong County in terms of forest carbon offset potential. *Spatial Information Research* 24, 1 (2016).

Peer-reviewed journal publications (*under review*)

- [U2] **Byeonghwa Jeong**; Junwoo Kim, Prediction of retail Gentrification Using a Data Science Approach; Case Study in Seoul, South Korea, *International Journal of Urban Sciences* (*under review*)
- [U1] Changwha Oh; Gaweon Shin; **Byeonghwa Jeong (corr. author)**. Exploring visitor-driven tourism themes: A bottom-up analysis using topic modelling of geotagged user-generated contents. *International Journal of Tourism Research* (*under review*).

Reports

- [R8] **Byeonghwa Jeong**; Viswadeep Lebakula; Marie Urban, PopGNN: Graph Neural Network-Based Flexible Future Population Forecasting Model, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States) (2026)
- [R7] **Byeonghwa Jeong**; Daniel Adams; Andrew Zimmer; Marie Urban, Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data: A Computer Vision-Based Approach, Oak Ridge National Laboratory (ORNL), Oak Ridge, TN (United States) (2026)
- [R6] Karen Chapple; Amir Forouhar; **Byeonghwa Jeong**; Stephanie Ortynsky, Assessing the Doom Loop, Lincoln institute of land policy (2025)
- [R5] Hyejin Joo; Sangheon Han; Eunjeong Jeong; **Byeonghwa Jeong**; Juhwan Song; Minyoung Noh, Big Data-Driven Strategies for Enhancing the 'Science City Daejeon' Brand: Focusing on the 2024 Korea Science Festival and Daejeon Science Festival, Daejeon Sejong Research Institute (2025)
- [R4] Hyejin Joo; Minyoung Noh; **Byeonghwa Jeong**, Basic Research for Citizen Participatory Urban Branding: Focusing on Human and Cultural Resources, Daejeon Sejong Research Institute (2024)
- [R3] Hyejin Joo; Sangheon Han; Seolmin Yoon; Jaeyeon Moon; **Byeonghwa Jeong**, Discover 'Regional Things' using Big Data Analysis and Civic Collaboration, Daejeon Sejong Research Institute (2023)
- [R2] Eunsun Lim; Youngju Lee; Bokyoung Lee; Yohan Jang; Jaesun Son; Kyungju Lee; Pyeongjik Lee; **Byeonghwa Jeong**; Hyungkwan Choi, Development and Utilization of Multiple Deprivation Index for Place-based Living Facilities Policy, Korea Research Institute for Human Settlements, (2021).
- [R1] Eunsun Lim; Youngju Lee; Bokyoung Lee; Pilsung Byun; Moonsoo Shin; **Byeonghwa Jeong**; Changhwa Oh; Gawon Shin; Kyungju Lee; Jongcheon Ahn; Hosang Lee; Jinho Park, National Territorial Monitoring Report in 2018, National Geographic Information Institute, (2019).

Work in preparation

- [P6] **Byeonghwa Jeong**; Sangkeun Lee, Aaron Myers, Dual-Structure Spatio-Temporal Graph Learning for Hurricane-Induced PowerOutage Forecasting, in preparation for submission to Association for the Advancement of Artificial Intelligence (AAAI) AI for Social Impact
- [P5] **Byeonghwa Jeong**; Viswadeep Lebakula; Marie Urban, PopGNN: Graph Neural Network-Based Flexible Future Population Forecasting Model, in preparation for submission to Association for the Advancement of Artificial Intelligence (AAAI) AI for Social Impact
- [P4] **Byeonghwa Jeong (corr. author)**; Joe Tuccillo; James Gaboardi; Building-Level Wealth Index Prediction Using DHS Survey Data and Machine Learning: A Flexible Multi-Scale Framework. in preparation for submission to *Computers, Environment and Urban Systems*.
- [P3] **Byeonghwa Jeong (corr. author)**; Andrew Zimmer; Daniel Adams; Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data: A Computer Vision-Based Approach. in preparation for submission to *ACM Knowledge Discovery and Data Mining (KDD)*.
- [P2] **Byeonghwa Jeong**; Viswadeep Lebakula; Brandyn Reynolds, Global level Livability Mask for Gridded Population Datasets. in preparation for submission to *Scientific Data*.
- [P1] Andrew Zimmer; Daniel Adams; **Byeonghwa Jeong**; Marie Urban. Framework for Rapid Updates to LandScan High-Resolution Gridded Population Datasets. in preparation for submission to *Scientific Reports*.

CONFERENCE PRESENTATION

- 2026 **Byeonghwa Jeong**; Joe Tuccillo; James Gaboardi, Building-Level Wealth Index Prediction Using DHS Data and Machine Learning, 2025 AAG(American Association of Geographers) Annual Meeting
- 2024 Young-Hoon Kim; **Byeonghwa Jeong**, Delineating commercial boundaries using an AI-based computer vision technique with open data in South Korea, AsiaCarto 2024
- 2024 **Byeonghwa Jeong**; Karen Chapple; Jeff Allen, Typology of Downtown Recovery at the Granular Scale, Geoinformatics2024

GRANTS

- [G3] A Study on the Evaluation and Characteristics of Coastal Area by Using Image Segmentation Model,
Korea Maritime Institute (KMI) \$2,885 (exchange rate 1\$ = 1,386.3 (Korea Won))
- Personal contribution: idea development, providing core data, writing and review
- [G2] Assessing the doom loop: Scenarios in four North American downtowns,
Lincoln Institute of Land Policy \$52,931 (exchange rate 1\$ = 1.37 CAD)
- Personal contribution: data curation, data collection, data preprocessing, explainable AI performing, writing proposal (some parts)
- [G1] MARSHALLING A ROBUST DOWNTOWN RECOVERY FOR OTTAWA,
Canadian Urban Institute \$109,514 (exchange rate 1\$ = 1.37 CAD)
- Personal contribution: data curation, data collection, data preprocessing, methodology selection, data analysis, visualization, writing proposal (some parts)

TEACHING AND SUPERVISION EXPERIENCE

Supervision

Undergraduate Students

- 2024 Christoper Kim (B.A in UC Berkeley, USA)
- 2024 Brandon Choi (B.A in UC Berkeley, USA)

Graduate Students

- 2024 Xiaoxin Zhou (M.S in Informatics, U of Toronto, Canada)
- 2024 Gaweon Shin (Ph.D in Public Administration, Yonsei University, South Korea)
- 2024 Changwha Oh (Ph.D in Geography, UT Knoxville, USA)

Guest Lecture

- 2025.3 The application of computer vision for delineating regional boundary, Center for territory monitoring, Korea Research Institute for Human Settlement (hosted by Dr Bokyeong Lee)
- 2024.8 The application of AI for social phenomenon prediction, Department of Geography Education, Korea National University of Education (hosted by Professor Younghoon Kim)
- 2023.6 Social phenomenon and spatial AI, Department of Geography, Kyungpook National University (hosted by Professor Jae Hyun Lee)

MANAGEMENT AND SERVICE

Conference and Event Management

- Member of the Organising Committee, International Conference on Geospatial Information Science, Spatially Enabled Society with AI and Digital Twin, 2019

Journal Reviewer

- International Journal of Geographical Information Science (IJGIS), Taylor & Francis, 2025
- Computers and Electrical Engineering, Elsevier, 2025
- Papers in Regional Science, Elsevier, 2026
- Spatial Information Research, Springer, 2026
- Transactions in GIS, Wiley, 2026

HONOURS AND PRIZES

- [A3] Awarded best paper for "Topic Modeling of the Transition of the Sense of Place in Urban Regeneration Area: A Case

Study of Daegu Bangcheon Market", Daegu Gyeongbuk Development Institute (Mar 2020)

[A2] Awarded an outstanding paper award by presenting 'Comparative Evaluation of Pixel Value Signature Between UN-A/R CDM Project Area in Goseong, South Korea and Goseong, North Korea', Conference on Geo-Spatial Information, KOREA Spatial Information Society (May 2015)

[A1] Awarded \$300 scholarship for the skilled ArcGIS person, June-Myeong Scholarship, GEO C&I Corp. (Mar 2013)

TECHNICAL SKILLS

- GIS analysis: ArcGIS, QGIS (Skilled)
- Computation Language: Python (Skilled)
 - Data query and management: Pandas, Geopandas, Numpy, PySpark
 - Machine Learning and Deep Learning: PyTorch, Torchvision, scikit-learn, Gensim
 - Traditional Statistics: Diverse regressions (e.g. Multivariate Multiple Regression), Likelihood ratio test, Diverse correlation test (e.g Pearson correlation coefficient), Feature engineering
 - Web-crawling: Selenium, BeautifulSoup, Request
 - Text-mining: TF-IDF, Word2vec, Latent Dirichlet Allocation
 - Large Language Models: Prompt engineering, fine-tuning
 - Network analysis: Networkx, UCINET
- Remote sensing: Google Earth Engine, Python-based raster analysis (Skilled)
- Cloud based server: Google Firebase and Colab (Skilled)
- Relational Database Management System: Mariadb (Have experienced)
- Statistic Software: R (Have experienced)